Chain Forward - Risk Assessment Summary

Executive conclusion

Recommendation: proceed with a controlled pilot rather than immediate full-scale rollout. The base-case economics generate $1.20M annual contribution after expected credit losses, but the $4.5M investment is recovered in 45 months rather than the 36-month target. Portfolio profitability is therefore highly sensitive to credit performance and operating assumptions.

Financial model Base case	$160K monthly revenue, $60K expected credit loss, $100K monthly contribution, $1.20M annual contribution, -20% 3-year contribution
Scenario analysis	Optimistic case reaches 30% 3-year ROI and 27.6-month payback; pessimistic performance falls to -86% ROI and 250-month payback
Customer segmentation	K-Means identified three behavioral groups. Cluster 2 combines high transaction value with substantially higher cash-flow volatility.
Default prediction	Logistic regression achieved 0.837 ROC-AUC. Cash-flow volatility was the strongest positive risk driver.
Model limitation	Default labels and transaction data used for the modeling demonstration are synthetic because customer-level labeled data was not provided.
Breakeven default rate	Credit losses alone break even at a ~16% monthly default rate; hitting the 36-month payback target requires default rates closer to ~3.5%.

Recommended risk controls

1. Use risk-based underwriting and pricing: segment borrowers using transaction behavior and default probability, while treating proposed pricing as illustrative until calibrated against real loss data.

2. Control concentration: set exposure limits by borrower, value chain/anchor and geography to reduce cascade risk.

3. Monitor early-warning indicators: rising cash-flow volatility, declining transaction frequency, falling repayment ratios, missed repayments, increasing utilization and repeat borrowing.

4. Optimize the default threshold: the 0.837 AUC is useful for ranking, but 0.308 recall at the current threshold means most actual defaults are missed; threshold selection should reflect the cost of false negatives.

5. Gate scale-up on evidence: validate the $100 EAD and 50% LGD assumptions, recalibrate the model with real repayment outcomes, and require the pilot to demonstrate a path to the 36-month payback target.

Chain Forward has a potentially viable risk-adjusted lending proposition, but the base case does not yet meet the investment hurdle. A controlled pilot with tight exposure limits, early-warning monitoring, risk-based pricing and rapid model recalibration is the appropriate next step.

Methodology

Profitability, breakeven, and return-on-capital were modeled analytically from stated and assumed unit economics. Segmentation used K-Means clustering on a synthetic MSME dataset (1,000 records) built around value-chain and behavioral features. Default prediction used logistic regression (AUC 0.837). The model identified cash-flow volatility as the strongest default driver, and transaction frequency and repayment ratio as protective.